



Cat

Microbiome

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Introduction



The purpose of this study is to look into the differences between the microbiomes found at two different body sites in cats: the rectum and the buccal areas.



I chose to study the microbiomes of cats because they are my favorite animals. I have a pet cat and it would be interesting to know a little bit about his microbiome.



I also love using cat memes and stickers when texting.



Methods

dada2: plotQualityProfile (Galaxy Version 1.38.0+galaxy0)

- This tool looks at the raw reads and provides a visual summary of the reads quality scores.

dada2: filterAndTrim (Galaxy Version 1.38.0+galaxy0)

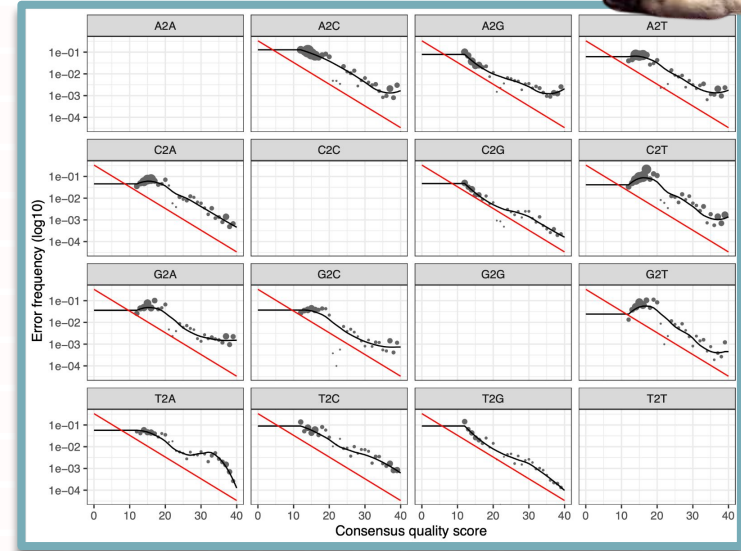
- This tool takes the raw reads and truncates them where the quality scores dropped below acceptable thresholds.

dada2: learnErrors (Galaxy Version 1.38.0+galaxy0)

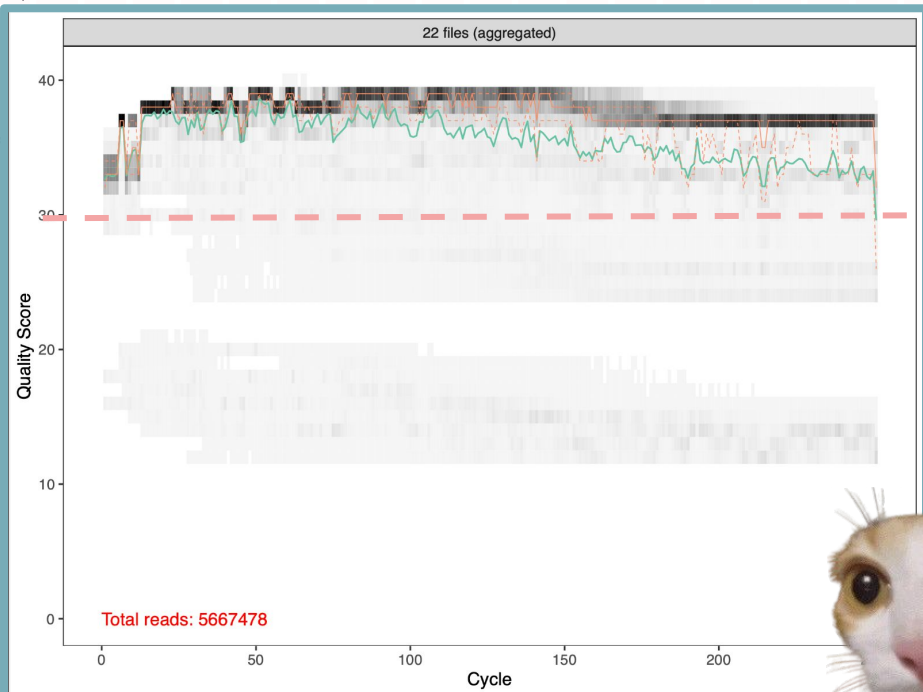
- This tool makes an error model for the sequencing data. It calculates the possibility of a possible nucleotide substitution.

dada2: dada (Galaxy Version 1.38.0+galaxy0)

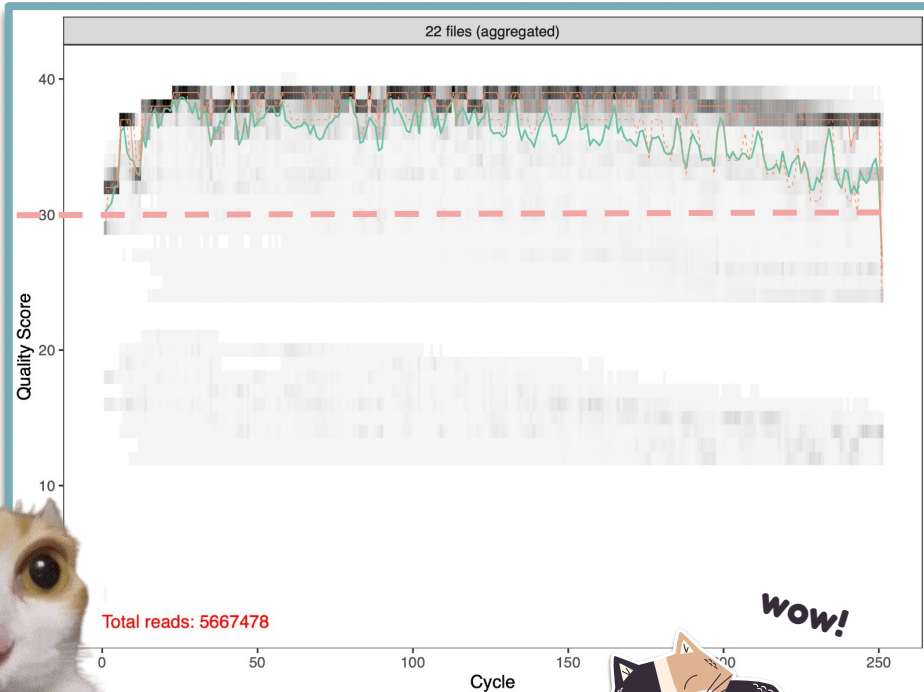
- This is a denoising tool. It uses the error model, fixes it, and gives you Amplicon Sequence Variants (ASVs).



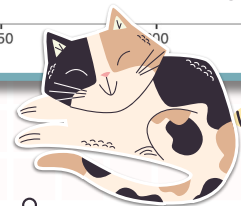
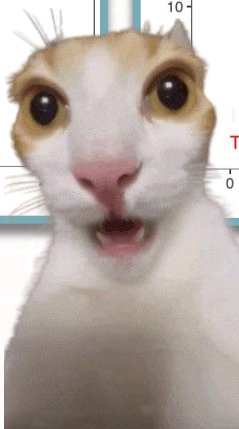
plotQualityProfile



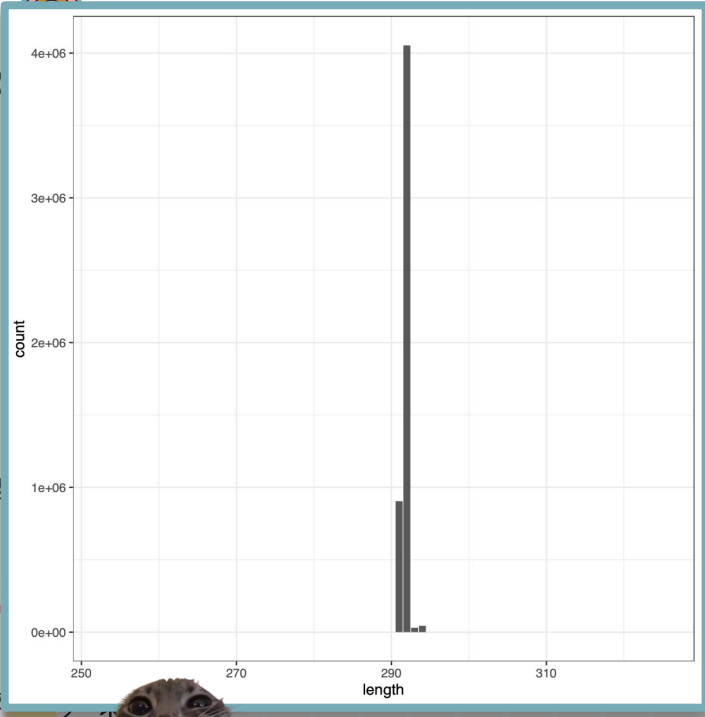
Forward reads



Reverse reads

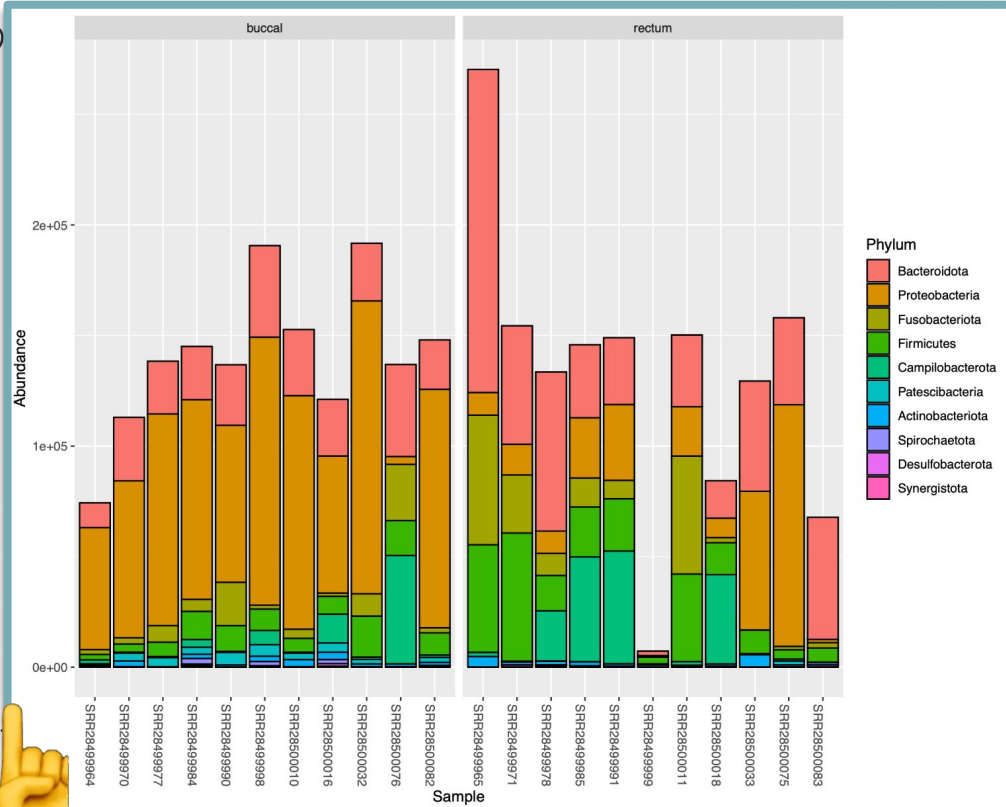


Methods Cont.



- ★ **dada2: makeSequenceTable** (Galaxy Version 1.38.0+galaxy0)
 - This tool takes the merged sequences and organizes them into a spreadsheet.
- ★ **dada2: removeBimeraDenovo** (Galaxy Version 1.38.0+galaxy0)
 - This tool gets rid of chimera. A chimera is a hybrid sequence that doesn't actually exist in nature. It is made during PCR.
- ★ **dada2: assignTaxonomy and addSpecies** (Galaxy Version 1.38.0+galaxy0)
 - This tool takes the clean ASV sequences and compares them to a database of known bacterial genomes. We used database Silva version 138.
- ★ **Create phyloseq object** (Galaxy Version 1.54.0+galaxy0)
 - This tool gets the removeBimera output, assignTaxonomy output, and metadata and puts them together in a single R data.

Bar Chart



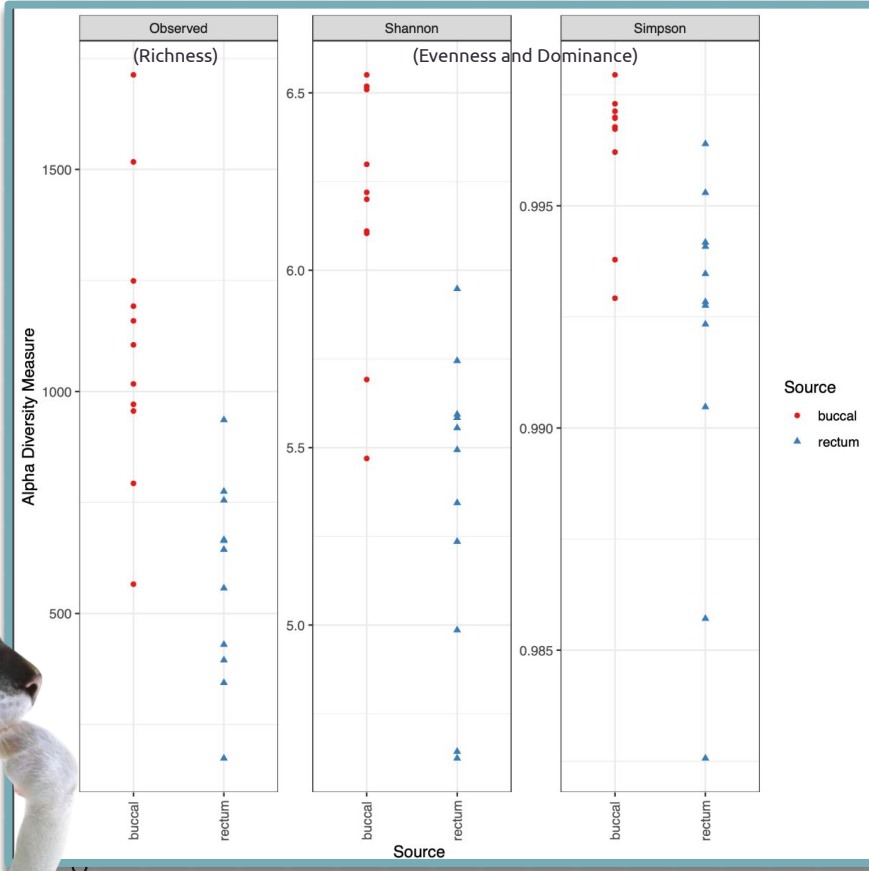
Proteobacteria is the most dominant bacteria in the buccal microbiome.

This is normal to see as it is a normal, prominent component of a healthy cat's microbiome.

Bacteroidota is a common bacteria among both microbiomes, but it is more abundant in the rectum microbiome.

It is also one of the dominant bacteria in a healthy cat's gastrointestinal microbiome. They are important in maintaining intestinal immunity.

Alpha Diversity Plot



Observed has a higher richness of ASVs in the buccal microbiome than the rectal microbiome.

The buccal samples sit high on the Shannon and Simpson plots.

The oral cavity has a diverse mix of microbes compared to the rectum area. This can be due to the cats constant exposure to new elements like food, toys, grooming their fur, etc.

Beta Diversity Plot

There is no overlapping between the two groups. Very good clustering between the rectum and fecal groups.

This is an outlier. Missed labeled swab? Cat licked rectum before swabbing?

Overlapping points means that samples are nearly identical.

